

The identity channel disproves a span criterion for symmetrizability

Conjecture 00000009543

October 8, 2026

Result. Containment of the identity map in the linear span of an arbitrarily varying quantum channel family does not imply symmetrizability. A singleton family containing the identity channel is a counterexample.

An actual quantum channel

Take $H = \mathbb{C}^2$, $S = \{s_0\}$, and let the only channel be

$$\mathcal{N}_{s_0} : M_2(\mathbb{C}) \longrightarrow M_2(\mathbb{C}), \quad \mathcal{N}_{s_0}(X) = X.$$

It is complex linear and trace preserving. It is completely positive, not merely positive: for every finite ancillary dimension n , the amplification $\text{id}_n \otimes \mathcal{N}_{s_0}$ is the identity on $M_n(M_2(\mathbb{C}))$, so it preserves every positive semidefinite matrix. Thus this is a genuine completely positive trace-preserving channel family. The identity belongs to its complex linear span because it is itself the sole member:

$$\text{id} \in \text{span}_{\mathbb{C}}\{\mathcal{N}_{s_0}\}.$$

The same observation holds for the real linear span if that convention is intended.

The necessary one-use symmetrization condition

The standard finite-ensemble condition at block length one asks that for every finite collection of density matrices $\rho_1, \dots, \rho_k$, there be probability distributions p_i on S such that

$$\sum_{s \in S} p_i(s) \mathcal{N}_s(\rho_j) = \sum_{s \in S} p_j(s) \mathcal{N}_s(\rho_i) \quad \text{for all } i, j. \quad (1)$$

The adversarial distribution is allowed to depend on the input state. In the usual all-block definition, this condition must hold at every block length; therefore failure at length one suffices to disprove symmetrizability. This is Definition 11 in the primary source cited below.

Consider the two density matrices

$$\rho_0 = \begin{pmatrix} 1 & 0 \\ 0 & 0 \end{pmatrix}, \quad \rho_1 = \begin{pmatrix} 0 & 0 \\ 0 & 1 \end{pmatrix}.$$

They are positive semidefinite, have trace one, and are different: their upper-left entries are one and zero respectively.

Failure of symmetrization

Every probability distribution p on the singleton S satisfies $p(s_0) = 1$. Hence, for every density matrix ρ ,

$$\sum_{s \in S} p(s) \mathcal{N}_s(\rho) = \rho.$$

Applied to the pair ρ_0, ρ_1 , equation (1) would require $\rho_1 = \rho_0$, a contradiction. Allowing separate input-dependent distributions does not help because a singleton has only one probability distribution.

Proposition 1. *There exists a finite family of completely positive trace-preserving maps whose linear span contains the identity, but which is not one-use symmetrizable and therefore is not symmetrizable in the all-block sense.*

Proof. The singleton identity-channel family constructed above has all the asserted properties. $\square$

Scope of the disproof

Both language versions assert the proposed span criterion for quantum AVC symmetrizability. The example refutes its sufficient direction. No nontriviality or lower bound on the size of S is stated; a singleton is a finite adversarial state set. The counterexample uses two genuine quantum input states and a quantum channel verified at every ancillary dimension.

This argument does not use capacity formulas, an entanglement-transmission convention, or an estimate of any symmetrization cost. It refutes the first claimed equivalence, which suffices to disprove the conjunction as written. In particular, it does not equate one-use symmetrizability with all-block symmetrizability; it uses only the necessary implication from all blocks to block length one.

Lean verification and reproduction

The supplied Lean 4.19.0 project pins Mathlib to

`c44e0c8ee63ca166450922a373c7409c5d26b00b`.

The formalization uses complex matrix linear maps, actual block amplification, Mathlib positive semidefiniteness, trace-one density matrices, and finite nonnegative weights summing to one. The full finite-ensemble one-use condition is quantified over every finite input list. The theorem `counterexample` combines complete positivity, trace preservation, span membership and failure of this necessary symmetrization condition. The final theorem negates the proposed sufficient implication directly. Run `lake build in lean/`; the source prints standard-only axiom audits.

Definition reference

R. Ahlswede, I. Bjelaković, H. Boche and J. Nötzel, *Entanglement Transmission over Arbitrarily Varying Quantum Channels*, ISIT 2010, Definition 11 and equation (18). Author-hosted conference paper. Only the definition is used; the counterexample above is self-contained.